

1. What is Gauss–Jordan Elimination?

Simple Idea

“We use row operations to convert a matrix into **identity matrix**.”

Why do we do this?

Because:

$$AX = B$$

We want:

$$X = ?$$

So we “undo” A using its inverse:

$$X = A^{-1}B$$

Gauss–Jordan is a method to FIND A^{-1}

2. Allowed Operations

You can only do **3 things to rows**:

1. Swap rows
2. Multiply a row by a number
3. Add/subtract rows

3. Core Idea of the Method

Start with:

$$[A \mid I]$$

Apply row operations until:

$$[I \mid A^{-1}]$$

4. What are we trying to achieve?

We want to make:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

This is called **Identity Matrix**

5. What is a Pivot?

A **pivot** is the “leading 1” in each row.

Example:

$$\begin{bmatrix} 1 & * & * \\ 0 & 1 & * \\ 0 & 0 & 1 \end{bmatrix}$$

6. Steps of Gauss–Jordan

- **Step 1**
 - Make pivot = 1
- **Step 2**
 - Make all values **below** pivot = 0
- **Step 3**
 - Make all values **above** pivot = 0

Repeat for all columns

Where You Are in the Process

After applying Gauss–Jordan steps:

You tried to convert

$$[A \mid I] \rightarrow [I \mid A^{-1}]$$

Now you **check what you got on the left side.**

Three Possible Outcomes (Special Cases)

Case 1: Perfect Identity (Best Case)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

What it means:

- Inverse exists
- Unique solution exists

So:

$$X = A^{-1}B$$

Case 2: Row becomes [0 0 0]

$$\begin{bmatrix} 1 & 0 & * \\ 0 & 1 & * \\ 0 & 0 & 0 \end{bmatrix}$$

What it means:

- Cannot form identity
- Inverse does NOT exist
- Matrix is **singular**

Interpretation

That row means:

$$0x + 0y + 0z = 0$$

No new information

One equation is redundant

Result

- Infinite solutions (if solving equations)
- No inverse (if finding inverse)

Case 3: Contradiction Row

$$[0 \quad 0 \quad 0 \mid 5]$$

Meaning:

$$0 = 5$$

Impossible

Result

- No solution at all
- System is inconsistent